

# Tianyi Zhang

404-422-2842 | tzhang708@gatech.edu | LinkedIn | GitHub | Google Scholar | tianyi-zhang-02.github.io

## EDUCATION

### Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, GA

Aug 2025 - May 2027

### Emory University

Bachelor of Science in Quantitative Sciences

Atlanta, GA

Aug 2021 - May 2025

## PROFESSIONAL EXPERIENCE

### LinkedIn Corporation

AI/ML Engineer Intern | Feed AI, LLM Retrieval

Mountain View, CA

May 2026 - Present

- Built a backend-agnostic **Adaptive User Representation Layer** that emits compact short-/long-term intents, lifecycle state, confidence, and evidence provenance as shared conditioning for embedding retrieval and generative recommendation.
- Built a production-scale **PySpark scorecard over 2.7B member-candidate pairs** and matched lifecycle evaluation; uncovered **+25.1% Recall@10** / **+26.8% NDCG@10** but **21.6% lower ILD** for Semi-cold versus Active members, informing model selection and lifecycle strategy that balance relevance with feed breadth.
- Improved distributed training and serving reliability through layered configuration, reproducible execution, train/serve parity checks, serving observability, and one-axis-at-a-time validation across representation, model capacity, retrieval backend, and lifecycle policy.

### Georgia Tech Efficient & Intelligent Computing Lab - NVIDIA Collaboration

Student Researcher | LLM Evaluation, Agentic AI

Atlanta, GA

Jul 2025 - Dec 2025

- Co-authored **AgentSuite (ICML 2026)**, auditing validity flaws in LLM-agent benchmarks; matched expert judgments at **0.79-0.87 F1** across 6 benchmarks and showed flaws affect **~23% of tasks**, reshuffling **63% of model rankings**.
- Built taxonomy-guided LLM-judge and rule-based structural detectors; generalized to unseen benchmarks without retuning and surfaced **100+** undocumented issues.

### Emory Graph Mining Lab

Machine Learning Researcher | Graph Learning, AI for Science

Atlanta, GA

Dec 2023 - May 2025

- First-authored a causal brain-connectivity method (**AIME 2025**) using Granger-directed graphs to improve clinical prediction across **~8K patients**.
- Co-authored a state-based Transformer (**IEEE EMBC 2025, oral**) modeling temporal transitions across latent functional-connectivity states.

## PUBLICATIONS

H. Suh\*, S. Lee\*, B. Ji\*, et al., including **T. Zhang**. "AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline." *ICML* 2026.

**T. Zhang**, K. Han, J. Nie, et al. "Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis." *AIME* 2025.

J. Nie, K. Han, **T. Zhang**, et al. "Brain Network State Transformer." *IEEE EMBC* 2025. (Oral)

## PROJECTS & OPEN SOURCE

**CoAct: Governance Substrate for Multi-Agent Coding** - Dependency-free Go binary for capability policies, locks, worktree isolation, merge gates, and append-only audit journals.

**NVIDIA NeMo RL: Open-Source Contributor** - Contributions to a scalable reinforcement-learning and LLM post-training library.

## TECHNICAL SKILLS

**Programming:** Python, Go, SQL, Java, R, C/C++ **ML/AI:** PyTorch, CUDA, Hugging Face, vLLM, SGLang, MLflow

**Infra/Data:** Spark/PySpark, Docker, AWS, GitHub Actions, Linux, PostgreSQL, MongoDB, Redis